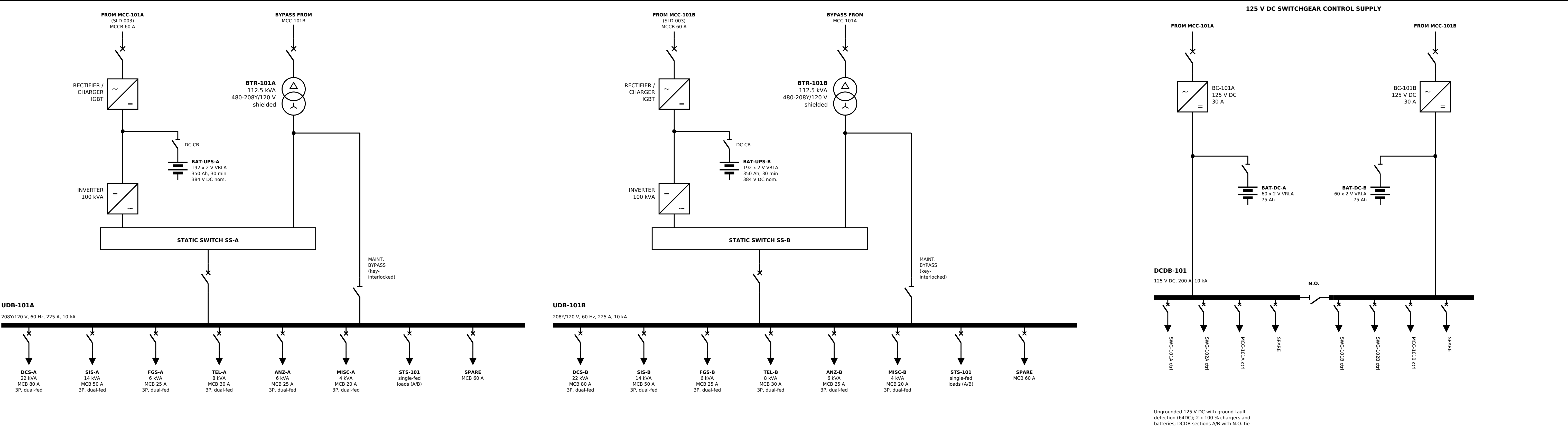




UPS / DC SIZING (detail in CFU-000-EL-CAL-001)	RESULT
Connected UPS load (DCS 22, SIS/BMS 14, F&G 6, telecom 8, analysers 6, misc 4)	60 kVA
Design load incl. 20 % future	72 kVA
UPS rating (design load <= 80 % of rating)	2 x 100 kVA
Battery power at design load (PF 0.9, inverter 94 %)	68.9 kW
Max. discharge current at end voltage 336 V (192 x 1.75 V)	205 A
Capacity = I x Kt 1.1 h (30 min) x aging 1.25 x margin 1.1	310 Ah -> 350 Ah
125 V DC duty: L1 60 A 1 min (trip) / L2 15 A 120 min standing / L3 40 A 1 min (close)	F = 49.6 Ah
125 V DC battery incl. aging 1.25, margin 1.10	68 Ah -> 75 Ah

CODE	UPS CONSUMER	kVA	A + B share
DCS	Distributed control system (controllers, I/O, servers, HMI in FAR)	22	11 + 11
SIS	Safety instrumented system incl. H-101/H-201 BMS, ESD solenoids	14	7 + 7
FGS	Fire & gas system incl. detectors/beacons	6	3 + 3
TEL	Telecom: PAGA, CCTV, LAN/fibre, radio	8	4 + 4
ANZ	Process analysers (AH-101)	6	3 + 3
MISC	Misc. instrument panels, MCC/IED networking, metering	4	2 + 2
TOTAL		60	

- NOTES**
- Emergency diesel generator NOT provided. Basis: two independent 13.8 kV feeders from the refinery main substation; the unit fails safe on total power loss (SIS de-energise-to-trip, fail-safe valves, heater BMS trip); no machinery needs post-trip power (no lube-oil / seal-oil systems, API 682 seals, steam ejectors). Essential loads (DCS, SIS/BMS, F&G, PAGA/telecom) on UPS 30 min; escape lighting self-contained 90 min. To be confirmed by HAZOP and the refinery power-system study.
 - UPS: 2 x 100 % dual-bus (UPS-101A -> UDB-101A, UPS-101B -> UDB-101B), each with own battery, static bypass and external maintenance bypass; 100 kVA each; input 480 V 3-ph 60 Hz (from MCC-101A/B); output 208Y/120 V 3-ph, 120 V 1-ph distribution, 60 Hz.
 - Consumers with redundant PSUs fed from both UDBs; single-fed loads via static transfer switch STS-101.
 - UPS battery: VRLA, 2 V cells, 20-yr design life, in ventilated battery room in SS-100; 192 cells, 350 Ah, 30 min at design load.
 - 125 V DC: 2 x 100 % chargers, 2 x 100 % batteries, DCDB-101 with A/B sections; batteries 75 Ah VRLA 60 cells, 2 h duty cycle (IEEE 485).
 - Battery rooms ventilated per IEEE 1635 / NFPA 70E; DC system ungrounded with ground-fault detection.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
SINGLE LINE DIAGRAM			
UPS & 125 V DC SYSTEMS			
DISCIPLINE	STAGE	SCALE	SHEET
ELECTRICAL	FEED	NTS	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-EL-SLD-004			A